

Mariadb : DBMS

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1. mariadb 설치

- mariadb-server : 서버 / 데이터가 저장되고 관리되는 '창고' 자체
- mariadb : 클라이언트 / 창고에 접속해 데이터를 요청하고 받아보는 '도구'

1) mariadb 설치

- Rocky Linux 9에서는 기본적으로 MariaDB가 EPEL(Extra Packages for Enterprise Linux) 리포지토리에 포함

`dnf install mariadb-server -y`

```
[root@localhost ~]# dnf install mariadb-server -y
```

```
Last metadata expiration check: 0:28:22 ago on Thu Sep 25 13:49:07 2025.
Dependencies resolved.
=====
Package                               Arch      Version                               Repository    Size
=====
Installing:
mariadb-server                        x86_64    3:10.5.27-1.el9_5.0.2               appstream    9.7 M
Installing dependencies:
mariadb                              x86_64    3:10.5.27-1.el9_5.0.2               appstream    1.6 M
mariadb-common                       x86_64    3:10.5.27-1.el9_5.0.2               appstream    27 k
mariadb-connector-c                  x86_64    3.2.6-1.el9_0                       appstream    195 k
mariadb-connector-c-config           noarch    3.2.6-1.el9_0                       appstream    9.8 k
mariadb-errmsg                       x86_64    3:10.5.27-1.el9_5.0.2               appstream    211 k
mysql-selinux                        noarch    1.0.14-1.el9_6                      appstream    36 k
perl-DBD-MariaDB                    x86_64    1.21-16.el9_0                       appstream    151 k
perl-DBI                             x86_64    1.643-9.el9                         appstream    700 k
Installing weak dependencies:
mariadb-backup                      x86_64    3:10.5.27-1.el9_5.0.2               appstream    6.5 M
mariadb-gssapi-server               x86_64    3:10.5.27-1.el9_5.0.2               appstream    14 k
mariadb-server-utils                x86_64    3:10.5.27-1.el9_5.0.2               appstream    210 k

Transaction Summary
=====
Install 12 Packages
```

2) mariadb 설치 확인

- `rpm -qa | grep mariadb`

```
[root@localhost ~]# rpm -qa | grep mariadb
mariadb-connector-c-config-3.2.6-1.el9_0.noarch
mariadb-common-10.5.27-1.el9_5.0.2.x86_64
mariadb-connector-c-3.2.6-1.el9_0.x86_64
mariadb-errmsg-10.5.27-1.el9_5.0.2.x86_64
mariadb-server-utils-10.5.27-1.el9_5.0.2.x86_64
mariadb-backup-10.5.27-1.el9_5.0.2.x86_64
mariadb-gssapi-server-10.5.27-1.el9_5.0.2.x86_64
mariadb-server-10.5.27-1.el9_5.0.2.x86_64
mariadb-10.5.27-1.el9_5.0.2.x86_64
```

- systemctl status mariadb

```
[root@localhost ~]# systemctl status mariadb
○ mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
```

4) mariadb 시작 및 확인

- systemctl start mariadb

- systemctl status mariadb

```
[root@localhost ~]# systemctl status mariadb
○ mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/

[root@localhost ~]# systemctl start mariadb
[root@localhost ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: active (running) since Thu 2025-09-25 14:21:23 KST; 2s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 990101 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 990123 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 990233 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
  Main PID: 990218 (mariabdb)
    Status: "Taking your SQL requests now..."
      Tasks: 14 (limit: 10857)
    Memory: 78.1M
      CPU: 628ms
    CGroup: /system.slice/mariadb.service
            └─990218 /usr/libexec/mariabdb --basedir=/usr
```

5) mariadb 초기화

- mysql_secure_installation

```
[root@localhost ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
```

```
Enter current password for root (enter for none):
```

- enter

```
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n]
```

- y

```
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n]
```

- y

```
Change the root password? [Y/n] y
New password:
```

- asd123!@

```
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n]
```

- y

```
Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.
```

```
Disallow root login remotely? [Y/n] █
```

-n

```
By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.
```

```
Remove test database and access to it? [Y/n] █
```

-y

```
- Dropping test database...  
... Success!  
- Removing privileges on test database...  
... Success!
```

```
Reloading the privilege tables will ensure that all changes made so far will take effect immediately.
```

```
Reload privilege tables now? [Y/n] █
```

-y

```
... Success!
```

```
Cleaning up...
```

```
All done! If you've completed all of the above steps, your MariaDB installation should now be secure.
```

```
Thanks for using MariaDB!
```

-완-

6) DBMS 에 접속 (root)

- mysql -uroot -p

- asd123!@

```
[root@localhost ~]# mysql -uroot -p  
Enter password:  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 15  
Server version: 10.5.27-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

7) 사용자 계정 생성

- create user 'park'@'%' identified by 'asd123!@';

```
MariaDB [(none)]> create user 'park'@'%' identified by 'asd123!@';  
Query OK, 0 rows affected (0.002 sec)
```

8) 데이터베이스(DB) 생성

- create database kong;

```
MariaDB [(none)]> create database park;  
Query OK, 1 row affected (0.001 sec)
```

9) (권한 위임) 'kong' DB에 대한 사용 권한을 'kong'@'%에게 위임

- grant all privileges on park.* to 'park'@'%';

```
MariaDB [(none)]> grant all privileges on park.* to 'park'@'%';  
Query OK, 0 rows affected (0.001 sec)
```

10) reload

- flush privileges;

```
MariaDB [(none)]> flush privileges;  
Query OK, 0 rows affected (0.001 sec)
```

11) 종료

- quit;

```
MariaDB [(none)]> quit;  
Bye
```

*) 명령어 모음

- mysql -upark -p : 만든 계정으로 로그인

```
[root@localhost ~]# mysql -upark -p  
Enter password:  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 16  
Server version: 10.5.27-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

- show databases; : 자신이 권한을 갖고 있는 데이터베이스(DB) 항목을 보여줌

```
MariaDB [(none)]> show databases;  
+-----+  
| Database |  
+-----+  
| information_schema |  
| park |  
+-----+  
2 rows in set (0.001 sec)
```

* 'park' DB 안에 어떤 내용이 있는지 궁금할 때

- use park;

```
MariaDB [(none)]> use park;  
Database changed  
MariaDB [park]> █
```

- show tables; : 어떤 데이터가 있는지 확인

```
MariaDB [park]> show tables;  
Empty set (0.000 sec)
```